

Homework Set HW14 due 5/3/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)

This is problem 1 Section 5.7 page 332.

Find c such that $\int_a^b f(x) dx = f(c)(b-a)$ for $f(x) = 4x^3$ on $[1, 2]$

 $c =$ _____

2.(1 pt)

This is problem 2 Section 5.7 page 332.

Find c such that $\int_a^b f(x) dx = f(c)(b-a)$ for $f(x) = x^2 + 4x + 1$ on $[0, 2]$

 $c =$ _____

3.(1 pt)

This is problem 17 Section 5.7 page 333.

Find the average value of $f(x) = x^2 - x + 1$ on $[-1, 2]$

Average Value = _____

4.(1 pt)

This is problem 17 Section 5.7 page 333.

Find the average value of $f(x) = x^3 - 3x^2$ on $[-2, 1]$

Average Value = _____

5.(1 pt)

This is problem 20 Section 5.7 page 333.

Find the average value of $f(x) = \frac{x}{2x+3}$ on $[0, 1]$

Average Value = _____

6.(1 pt)

This is problem 22 Section 5.7 page 333.

Find the average value of $f(x) = 2\sin(x) - \cos(x)$ on $[0, \pi/2]$

Average Value = _____

7.(1 pt)

This is problem 24 Section 5.7 page 333.

Find the average value of $f(x) = \sqrt[3]{1-x}$ on $[-7, 0]$

Average Value = _____

8.(1 pt)

This is problem 26 Section 5.7 page 333.

Find the average value of $f(x) = x\sqrt{2x^2+7}$ on $[0, 1]$

Average Value = _____

9.(1 pt)

This is problem 33 Section 5.7 page 333.

Records indicate that t hours past midnight, the temperature at the local airport was

$$f(t) = -0.1t^2 + t + 50$$

degrees Fahrenheit. What was the average temperature at the airport between 9 : 00 A.M. and noon?

Average Value = _____ degrees Fahrenheit

10.(1 pt)

This is problem 34 Section 5.7 page 333.

Suppose a study indicates that t years from now, the level of carbon dioxide in the air of a certain city will be

$$L(t) = te^{-.01t^2}$$

parts per million (ppm) for $0 \leq t \leq 20$?

(a) Find the average level of carbon dioxide in the first three years _____ ppm

(b) Find the two times $t_1 < t_2$ (in years) that the average level of carbon dioxide actually occurs. (give answers to the nearest one hundredth of a year)

 $t_1 =$ _____ $t_2 =$ _____

11.(1 pt)

This is problem 3 Section 5.8 page 339.

Approximate the integral $\int_0^1 \frac{dx}{1+x^2}$ using $n = 4$ intervals and using the trapezoid rule and Simpson's rule. (Give answers to within 4 digit accuracy.)

(a) The trapezoid rule : _____

(b) Simpson's rule : _____

12.(1 pt)

This is problem 6 Section 5.8 page 339.

Approximate the integral $\int_0^2 x \cos(x) dx$ using $n = 6$ intervals and using the trapezoid rule and Simpson's rule. (Give answers to within 4 digit accuracy.)

(a) The trapezoid rule : _____

(b) Simpson's rule : _____

13.(1 pt)

This is problem 8 Section 5.8 page 339.

Approximate the integral $\int_1^2 \ln(x) dx$ using $n = 6$ intervals and using the trapezoid rule and Simpson's rule. (Give answers to within 4 digit accuracy.)

(a) The trapezoid rule : _____

(b) Simpson's rule : _____

14.(1 pt)

This is problem 20 Section 5.8 page 339.

Determine how many subintervals are required to guarantee accuracy to within .00005 for the integral $\int_{-1}^4 (x^3 + 2x^2 + 1) dx$ using the trapezoid rule and Simpson's rule.

(a) The trapezoid rule : $N =$ _____(b) Simpson's rule : $N =$ _____

15.(1 pt)

This is problem 22 Section 5.8 page 339.

Determine how many subintervals are required to guarantee accuracy to within .00005 for the integral $\int_0^2 \cos(x) dx$ using the trapezoid rule and Simpson's rule.

(a) The trapezoid rule : $N =$ _____

(b) Simpson's rule : $N =$ _____